

4.2.1 Systems in the human body

Systems in the human body can be studied at macroscopic, microscopic and molecular scales. The study of respiration helps to account for the need for exchange surfaces in multicellular organisms, illustrated by the human circulatory system. The study of the digestive system focuses on the chemical changes to the main nutrients in the diet. Finally, examples of the way that body systems are controlled is illustrated with reference to the nervous system and the endocrine system. In [Lifestyle and health](#) (page 50) the importance of the endocrine system is further illustrated in the context of lifestyle and health.

There are two required practicals: one is using reagents to test for a range of carbohydrates, lipids and proteins, the other is an investigation into the effect of a factor on human reaction times, which can be related to road safety in [Stopping distances](#) (page 115).

4.2.1.1 Respiration

GCSE science subject content	Details of the science content	Scientific, practical and mathematical skills
Describe cellular respiration as an exothermic reaction which is continuously occurring in all living cells.	Respiration in cells can take place aerobically (using oxygen) or anaerobically (without oxygen). Aerobic respiration is an exothermic reaction that can be represented by word and symbol equations. glucose + oxygen → carbon dioxide + water An exothermic reaction is one that transfers energy to its surroundings. Organisms need energy for: • chemical reactions to build larger molecules • movement • keeping warm.	WS 1.2 (HT only) Write a balanced symbol equation for respiration, given the formula of glucose.
Compare the processes of aerobic and anaerobic respiration.	Anaerobic respiration in muscles is also exothermic but it gives out less energy. It is represented by the word equation: glucose → lactic acid Because the oxidation of glucose is incomplete in anaerobic respiration much less energy is given out than in aerobic respiration. If insufficient oxygen is supplied, anaerobic respiration takes place in muscles. The incomplete oxidation of glucose causes a build-up of lactic acid and creates an oxygen debt. Oxygen debt is the amount of extra oxygen the body needs after exercise to react with the accumulated lactic acid and remove it from the cells.	

4.2.1.2 Exchange surfaces

GCSE science subject content	Details of the science content	Scientific, practical and mathematical skills
Explain the need for exchange surfaces and a transport system in multicellular organisms in terms of surface area:volume ratio.	A single-celled organism has a relatively large surface area:volume ratio. The tissues of a multicellular organism consist of cells with a similar structure and function. Organs, such as the heart and lungs, are made of tissues. One organ may consist of several tissues. Organ systems, such as the circulatory system, are groups of organs that perform a particular function. In multicellular organisms many organ systems are specialised for exchanging materials. The effectiveness of an exchange surface is increased by: • having a large surface area • a membrane that is thin, to provide a short diffusion path • (in animals) having an efficient blood supply • (in animals, for gaseous exchange) being ventilated.	MS 1c Calculate and compare surface area:volume ratios.

4.2.1.3 The human circulatory system

GCSE science subject content	Details of the science content	Scientific, practical and mathematical skills
Describe the human circulatory system, including the relationship with the gaseous exchange system, and explain how the structure of the heart and the blood vessels are adapted to their functions. Describe some of the substances transported into and out of a range of organisms in terms of the requirements of those organisms, to include oxygen, carbon dioxide and dissolved food molecules.	The heart is a muscular organ that pumps blood around the body in a dual circulatory system. The right ventricle pumps blood to the lungs, where gas exchange takes place. The left ventricle pumps blood around the rest of the body. Valves prevent the blood from flowing back from the ventricles to the atria. Knowledge of the names of the heart valves is not required. Blood vessels associated with the heart include the aorta, vena cava, pulmonary artery, pulmonary vein and coronary arteries. Gas exchange takes place in the lungs. Important features of the lungs are the trachea, bronchi, alveoli and the capillary network surrounding the alveoli. The alveoli have the specialised surfaces for gas exchange between air and the blood. The natural resting heart rate is controlled by a group of cells that act as a pacemaker, located in the right atrium. Artificial pacemakers are electrical devices used to correct irregularities in the heart rate. The body contains three different types of blood vessel: • arteries • veins • capillaries.	MS 1a, 1c Use simple compound measures such as rate. MS 1a, 1c Carry out rate calculations.

4.2.1.4 Blood cells

GCSE science subject content	Details of the science content	Scientific, practical and mathematical skills
Explain how red blood cells, white blood cells, platelets and plasma are adapted to their functions in the blood.	Blood is a tissue consisting of plasma, in which are suspended: • red blood cells • white blood cells • platelets.	WS 3.5 Identify different types of blood cells in a photograph or diagram.

4.2.1.5 The human digestive system

GCSE science subject content	Details of the science content	Scientific, practical and mathematical skills
Explain the importance of sugars, amino acids, fatty acids and glycerol in the synthesis and breakdown of carbohydrates, lipids and proteins. Describe some of the substances transported into and out of a range of organisms in terms of the requirements of those organisms, to include dissolved food molecules and urea.	The digestive system uses enzymes to break down large molecules in food into small soluble molecules that can be absorbed into the blood through the walls of the gut. The blood carries the small molecules to the cells of the body where they can be used for respiration or to make the new large molecules that the cells need as reserves of energy or for growth and repair. Starch is a carbohydrate. Its molecules consist of a long chain of glucose molecules. Digestion by carbohydrase enzymes breaks down insoluble starch to water-soluble glucose. Cells use glucose during respiration. Lipids are fats and oils. Digestion by lipase enzymes breaks down lipids to glycerol and fatty acids. Cells reform fats from the fatty acids and glycerol molecules. Fats are stored as a source of energy because cells can break them down and use them in respiration. Proteins are long-chain molecules made up of many amino acids linked together. Digestion by protease enzymes breaks down proteins to amino acids. Cells use amino acids to make new proteins. The liver breaks down unwanted amino acids to urea, which is then carried by the blood to the kidneys. The kidneys excrete urea in solution as urine.	

Required practical activity 7: use qualitative reagents to test for a range of carbohydrates, lipids and proteins. To include: Benedict's test for sugars, iodine test for starch and Biuret reagent for protein.

AT skills covered by this practical activity: biology AT 2.

This practical activity also provides opportunities to develop WS and MS. Details of all skills are given in [Key opportunities for skills development](#) (page 166).

4.2.1.6 The human nervous system

GCSE science subject content	Details of the science content	Scientific, practical and mathematical skills
Explain how the structure of the nervous system (including the central nervous system, sensory and motor neurones and sensory receptors) is adapted to its functions.	The nervous system enables humans to react to their surroundings and to coordinate their behaviour. Information from receptors passes along cells (neurones) as impulses to the central nervous system, or CNS (the brain or the spinal cord). The CNS coordinates the response of effectors which may be muscles contracting or glands secreting hormones. stimulus → receptor → coordinator → effector → response	
Explain how the structure of a reflex arc is related to its function.	Reflex actions are automatic and rapid; they do not involve the conscious part of the brain. An example of a simple reflex action is the pain withdrawal reflex. This can be explained in terms of a reflex arc. Sensory neurones carry impulses from receptors to the spinal cord and brain. Relay neurones carry impulses within the CNS. Motor neurones carry impulses from the CNS to effectors. Where two neurones meet, there is a tiny gap called a synapse. Impulses cross this gap using chemicals.	
Explain methods of measuring human reaction times and recall typical results.	Reaction times vary from person to person. Typical values range from 0.3s to 0.9s.	This topic links with Stopping distances (page 115).

Required practical activity 8: plan and carry out an investigation into the effect of a factor on human reaction time.

AT skills covered by this practical activity: biology AT 1, 3 and 4.

This practical activity also provides opportunities to develop WS and MS. Details of all skills are given in [Key opportunities for skills development](#) (page 167).

4.2.1.7 The human endocrine system

GCSE science subject content	Details of the science content	Scientific, practical and mathematical skills
Describe the principles of hormonal coordination and control by the human endocrine system.	The endocrine system is composed of glands that secrete hormones directly into the bloodstream. Hormones are large molecules. The blood carries the hormone to a target organ where it produces an effect. Compared to the nervous system the effects are slower but act for longer. The pituitary gland in the brain is a 'master gland'. It secretes several hormones that act on other glands to stimulate other hormones to be released.	
(HT only) Explain the roles of thyroxine and adrenaline in the body including thyroxine as an example of a negative feedback system.	(HT only) Adrenaline is produced by the adrenal gland. It boosts the delivery of oxygen and glucose to the brain and muscles and prepares the body for 'flight or fight'. (HT only) Thyroxine from the thyroid gland stimulates the basal metabolic rate. It plays an important role in growth and development. (HT only) The control of thyroxine levels involves negative feedback. Negative feedback tends to stabilise a system. Any change in the system leads to a response that tends to reverse the change.	WS1.2, MS 2c (HT only) Interpret and explain simple diagrams of negative feedback control.